

```
goal = [  
    [1,2,3],  
    [4,5,6],  
    [7,8,0]  
]  
  
start = [  
    [1,2,3],  
    [4,0,6],  
    [7,5,8]  
]  
  
def manhattan(board):  
    distance = 0  
  
    for i in range(3):  
        for j in range(3):  
            value = board[i][j]  
  
            if value == 0:  
                continue  
  
            x = (value-1)//3  
            y = (value-1)%3  
  
            distance += abs(i-x)+abs(j-y)  
  
    return distance  
  
print("Heuristic =", manhattan(start))
```